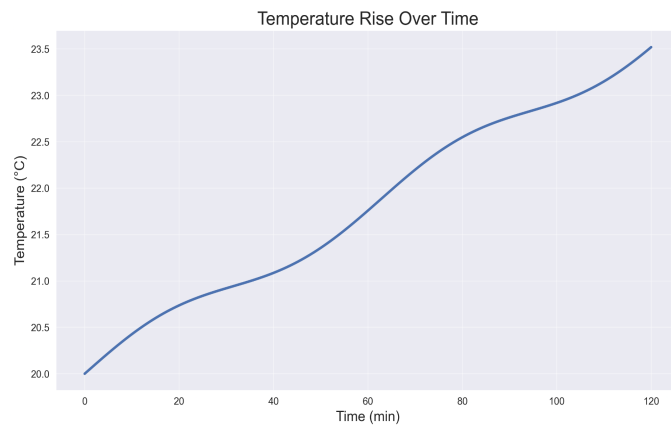
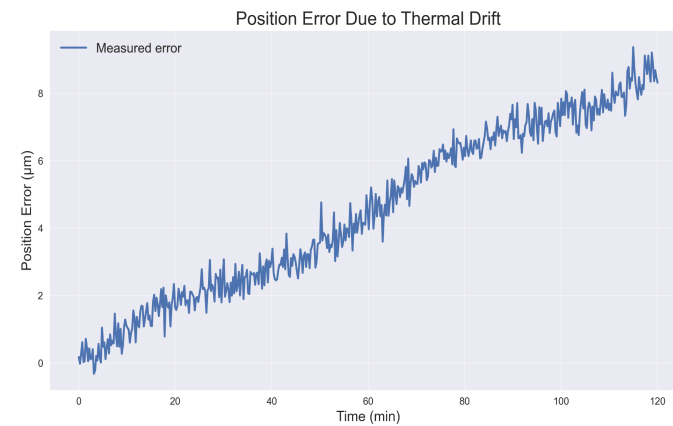


Thermal Drift Analysis in Precision Machine Systems

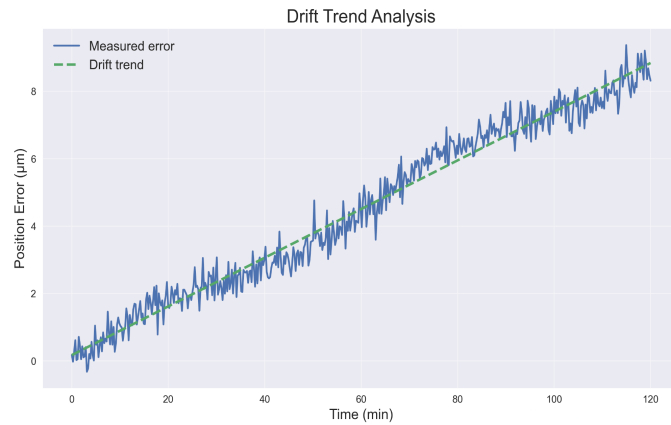
Simulation Study of Temperature Induced Position Drift and Compensation Performance



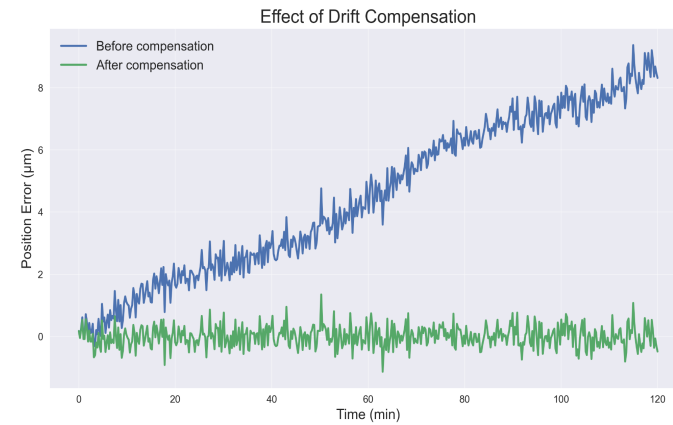
1. Machine Warm Up: Temperature increases gradually during machine warm up.



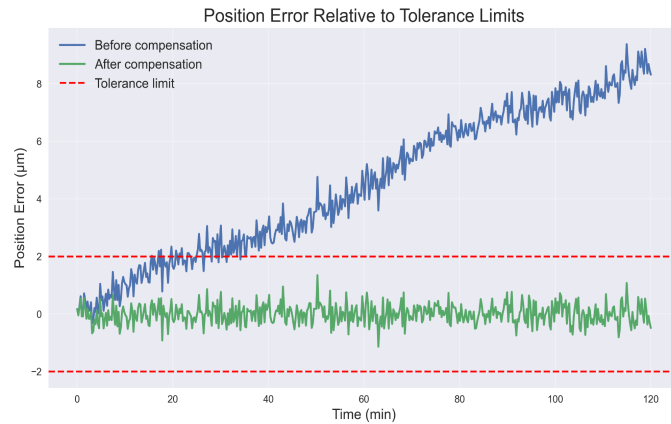
2. Error Development: Thermal expansion introduces increasing positioning error.



3. Drift Trend: A clear systematic drift trend appears in the positioning error.



4. Compensation Strategy: Temperature based compensation significantly reduces drift.



5. Tolerance Evaluation: Compensated error remains largely within acceptable limits.